This is the authors' version of the following article: Lee, H., & Lee, J. H. (2023). Interpreting the effectiveness of academic English writing programmes in higher education: A meta-analysis. *Oxford Review of Education*, *49*(3), 281–300. The version of record is available at https://doi.org/10.1080/03054985.2022.2072284. This version differs from the published article in layout and pagination; please cite the published version.

Interpreting the effectiveness of academic English writing programmes in higher education: A meta-analysis

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Supplemental data for this article can be accessed online at https://doi.org/10.1080/03054985.2022.2072284

Abstract

This study synthesised previous literature focused on college English for Academic Purposes (EAP) programmes for writing. A meta-analysis of identified longitudinal studies (14 samples; 175 effect sizes; $N = 602$) revealed that the students improved their English writing through EAP programmes ($d = .45$, $SE = .07$, $p < .001$). By computing effect size mean estimates from ranges for the 25th, 50th, and 75th percentiles, we proposed effect size interpretation benchmarks for this subdomain of 0.3, 0.5, and 0.7 for small, medium, and large effects, respectively, for interpreting the improvements in writing through college EAP programmes. The meta-regression revealed that the students with low English proficiency improved significantly more than those with higher proficiency. Further, the programmes had a small effect in improving accuracy and complexity dimensions of writing. Different features of programmes and contexts have also been found to influence the degree of the improvements in complex ways.

Keywords: college Writing; interpretation benchmarks; English for academic purposes; English writing; meta-analysis

Introduction

English for Academic Purposes (EAP) courses comprise various programmes, curricula, and activities designed to enhance students' skills related to the comprehension and use of English in academic discourses for their studying of specific disciplines and for successful use in their careers. While these components vary across programmes, including listening to academic lectures (Rodgers & Webb, 2016), building academic and disciplinary vocabulary (Coxhead,

2016), and gaining fluency in spoken interaction (Basturkmen, 2016), writing has arguably gained most attention from instructors, researchers, and students, because it is ‘often the only means by which students’ content knowledge is assessed in a large number of disciplines’ (Mazgutova & Kormos, 2015, p. 3). English academic writing skills have been suggested to be essential to successful academic achievement (Graham et al., 2016).

Owing to its importance, a considerable number of EAP programmes, in particular postgraduate-level ones, ‘aim to develop the learners’ writing skills, including development of more sophisticated and accurate levels of academic language use’ (Storch & Tapper, 2009, pp. 207–208). EAP programmes may be particularly important for incoming higher education students because they often do not have adequate EAP writing skills (National Center for Education Statistics [NCES], 2012). In these programmes, instructors focus on English academic writing, give feedback on students’ writing, organise conferences on academic writing, and make various efforts to develop students’ English academic writing skills.

Some researchers (e.g. Crossley & McNamara, 2014; Crosthwaite, 2016; Henry & Roseberry, 1998; Shaw & Liu, 1998) have addressed the extent to which these college EAP programmes could aid in improving various linguistic dimensions of students’ writing, with the students usually being registered in such programmes at their own institutions. In general, they have adopted a pre test–posttest design, under which students’ academic writing is collected at two different time points to gauge the progress and assess complexity, accuracy, fluency, and other measures. As each study has only provided partial evidence regarding the development of different dimensions of writing, a synthesis of their findings would reveal the extent to which these programmes could enhance students’ English writing.

To this end, we conducted a meta-analysis to synthesise previous longitudinal studies on college EAP programmes that examined diverse measures of writing. Our goals were three-fold: (1) to calculate an overall average effect size (ES) related to students’ improvements in English writing through EAP programmes, (2) to provide domain-specific ES interpretation benchmarks for their improvements in English writing through such programmes, and (3) to examine the roles of target moderator variables (i.e. students’ English proficiency, target linguistic dimensions of writing, instructional conditions, duration of the EAP programme, and contexts).

Literature review

Theoretical background to the success of EAP programmes for writing

Despite some variations across programmes, EAP programmes are primarily designed to improve students' ability to communicate successfully in academic contexts, with writing at the centre of the curriculum (Mazgutova & Kormos, 2015). One of the variations lies in the level of specificity (Hyland, 2016), with some programmes being more customised to specific disciplines and others addressing generic features of academic English (e.g. rhetoric, organisation, formality, effectiveness of communication) transferable across different disciplines. Blue (1988) terms the latter 'English for general academic purposes (EGAP)', designed for students in the early stages of their academic study. It should be noted that all the studies included in the current meta-analysis are based on EGAP, and included EFL or ESL students, who are the general target audience of college EAP programmes (Tardy & Jwa, 2016).

The effectiveness of EAP programmes for writing can be theoretically supported, as these, albeit with some differences in the programme features, have adopted common theoretical frameworks and pedagogical approaches, which are discussed below. First, EAP programmes have generally adopted a task-based approach (Skehan, 1996), through which students are given a wide range of writing tasks, ranging from essays on general academic topics, critical literature reviews of relevant fields, to proposals for their research. Practising such writing is a fundamental element of the curriculum and assessment in EAP programmes. In addition, these programmes have adopted the genre approach (Hyland, 2007, 2008), in which students' writing is expected to achieve a particular purpose of effective communication with the readership, helping them grasp the essential elements in the academic content. Furthermore, as writing is imperative, providing feedback on students' writing, particularly vocabulary and syntax (Bitchener & Ferris, 2012; Ferris, 2002; Leki, 2006), is another fundamental component.

Manchón (2016) suggested that students' engagement in writing tasks and associated feedback processing could lead to significant second language (L2) development and that engaging in writing tasks could enhance students' fluency 'as a result of the conditions for communicative practice that writing academic texts favors' (p. 146). Furthermore, Second Language Acquisition frameworks such as focus-on-form (i.e. feedback on linguistic errors in EAP programmes is contextualised, cf., Crosthwaite, 2016; Long & Robinson, 1998) and the noticing hypothesis (learners' attention is shifted towards the errors highlighted by their instructors in the process of revision, cf., Ishikawa, 1995; Schmidt, 2001) offer theoretical support for the idea that feedback processing could contribute to the registration and

consolidation of L2 knowledge. Despite such supports, Manchón (2016) notes that more research is needed to support the effects of writing on L2 developments in academic settings. With James' (2014) qualitative systematic review of learning transfer in EAP contexts being, to the best of the authors' knowledge, the only available research synthesis on EAP contexts, the present meta-analysis has a significant potential to add to the evidence regarding the effectiveness of EAP programmes for writing.

Potential moderators of the overall success of EAP programmes for writing

In this section, we introduce the identified moderators for the present meta-analysis (i.e. learners' English proficiency, target linguistic dimensions of writing, instructional conditions, duration, and contexts), along with rationales for including them.

Learners' English proficiency. Most college EAP programmes have been designed to cater to the needs of the students at rather earlier stages of their academic studies, although they are also open to those with higher levels of English proficiency. Indeed, primary studies have included learners with diverse levels of English proficiency (e.g. low in Ishikawa, 1995; intermediate in Crossley & McNamara, 2014; high in Alghizzi & Alshahrani, 2020), with some (e.g. Mazgutova & Kormos, 2015) revealing that the proficiency level influenced the effectiveness of college EAP programmes. Thus, in the present study, learners' English proficiency is included as one of the moderators.

Target linguistic dimensions. The primary studies on this issue have focused on different linguistic dimensions of writing (see Table 5 for details). For example, a majority have adopted the complexity, accuracy, and fluency (CAF) framework (Ellis, 2003; Ellis & Barkhuizen, 2005; Skehan, 1998; Wolfe-Quintero et al., 1998) in operationalising students' writing performance or proficiency. Among these, (linguistic) complexity is characterised as 'the size, elaborateness, richness, and diversity of the learner's linguistic L2 system' (Housen & Kuiken, 2009, p. 464), although it has been defined in several different ways with different types of complexity having been proposed (e.g. linguistic complexity, propositional complexity, discourse-interactional complexity) (see Bulté & Housen, 2014 for review). Accuracy is operationalised as the extent to which one's written output deviates from a target standard (or norm), and fluency as smoothness of writing or amount of written production per a given period of time. In addition to the CAF framework, some studies have employed holistic assessments, which are based on 'reading for an individual impression of the quality of the writing' (Hamp-Lyons, 1990, p. 79), or examined the levels of cohesion (Crossley et al., 2016) and academic register (Crosthwaite, 2016) as measures of the development of students' writing through EAP programmes. Thus, to synthesise the findings of the primary studies with different linguistic foci, target linguistic dimension was included as another moderator.

Instructional conditions. Among the primary studies on this issue, some included both a traditional (i.e. business-as-usual) condition and an interventional one. Whereas the former may refer to a typical condition that has the features of general EAP programmes (see *Theoretical background to the success of EAP programmes for writing* section), the latter refer to more context-specific enhanced conditions, such as genre-based teaching in Henry and Roseberry (1998), extensive reading-based in Park (2015), and revision-enhanced condition in Polio et al. (1998). To examine the role of such conditions, we included a moderator termed 'instructional conditions'.

Duration. Another variation among the methodological features relates to the duration of the EAP studies on the development of writing. We found that some have collected the data over the period of one semester (e.g. Alghizzi & Alshahrani, 2020; Park, 2015), while others for less (e.g. three weeks in Henry & Roseberry, 1998, four weeks in Mazgutova & Kormos, 2015). Accordingly, the moderator termed 'duration' was coded and included in our analysis.

Context. EAP programmes have targeted both EFL (i.e. where English is not the primary language spoken outside the classroom) and ESL contexts (i.e. English is the dominant language in the target community; Tardy & Jwa, 2016). The former includes Saudi Arabia (Alghizzi & Alshahrani, 2020) and Japan (Ishikawa, 1995), whereas the latter includes the UK (Mazgutova & Kormos, 2015) and the US (Crossley & McNamara, 2014). This difference was operationalised as the moderator termed 'context'.

Aims and research questions

The present meta-analysis is motivated by the lack of synthetic effort that has addressed the important issue of the extent to which the EAP programmes could improve students' English writing. Thus, we aggregate the results of the existing primary studies on this issue and provide an overall estimate of the effectiveness of EAP programmes for English writing. We further sought to establish informative ES interpretation benchmarks for college EAP programmes based on our results, which those concerned with college EAP programmes could use in the evaluation of the success of their own programmes. Such effort may correspond to Plonsky and Oswald's (2014) suggestion that more domain-specific ES interpretation benchmarks (rather than one-size-fits-all standards) are needed to interpret the findings of an individual study within that domain more accurately. Finally, we identified a range of potential moderators as reviewed above and their effects to provide evidence-based suggestions for those involved in EAP programmes to effectively manage their programmes for English writing.

In view of the aforementioned aims, we addressed the following three research questions:

(1) To what extent do college EAP programmes improve students' English writing?

(2) What are the ES interpretation benchmarks for the improvements in English writing through college EAP programmes?

(3) To what extent do the selected moderator variables influence the degree of improvements in English writing through college EAP programmes?

Methods

Literature search

We used three keywords in combination with 'English for academic purposes', namely, 'development', 'impact', and 'effectiveness', across three databases, ProQuest (including Dissertation abstracts Databases), Scopus, and Web of Science, and relevant journals, including *Applied Linguistics, Assessing Writing, ELT Journal, English for Academic Purposes, Journal of Academic Writing, Journal of English for Academic Purposes, Journal of Second Language Writing, Language Learning, Reading & Writing,* and *The Modern Language Journal*. We also conducted a search of references in articles and a backward search of articles using Google Scholar.

When the identified studies were gathered, a total of 2,186 studies had 636 duplicates. To include only EAP writing empirical research, we first reviewed titles and abstracts of 1,550 studies and then analysed the PDFs of the remaining studies according to our seven inclusion criteria (see Table 1): a study should (1) feature college students, (2) be about a writing programme, (3) be related to EAP, (4) be an empirical study, (5) be based on a quantitative approach, (6) focus on student learning outcomes, and (7) report relevant descriptive statistics for ES calculations.

We excluded a total of 1,536 studies for the following reasons: (1) 676 studies were not about college students, (2) 352 were not about a writing programme, (3) 249 were not related to EAP, (4) 49 were not empirical studies (e.g. review, conceptual papers), (5) 77 were qualitative studies (e.g. case studies, ethnographic studies), and (6) 133 were not focused on student learning outcomes. As a result, 14 studies remained. It should be noted that all the identified studies implemented a one group pre test–posttest design to examine the extent to which the students improved in their English writing (rather than including comparison groups, which study in non-EAP conditions), probably because colleges cannot arbitrarily deprive their low proficiency students of the opportunity to take EAP courses in a randomised control trial for a research purpose. The entire literature search process was conducted by the

authors of the present study and any disagreements were resolved through a series of discussions. A preferred reporting items for systematic reviews and meta-analyses (PRISMA; Page et al., 2021) flow chart for the literature search procedure is presented in Figure 1.

Table 1. Inclusion criteria and the results of literature selection.

Inclusion criteria	Excluded studies	Remaining studies
(1) a study should feature college students	676	874
(2) a study should be about a writing programme	352	522
(3) a study should be related to EAP	249	273
(4) a study should be an empirical study	49	224
(5) a study should be based on a quantitative approach	77	147
(6) a study should focus on student learning outcomes	133	14
(7) a study should report relevant descriptive statistics for ES calculations	0	14

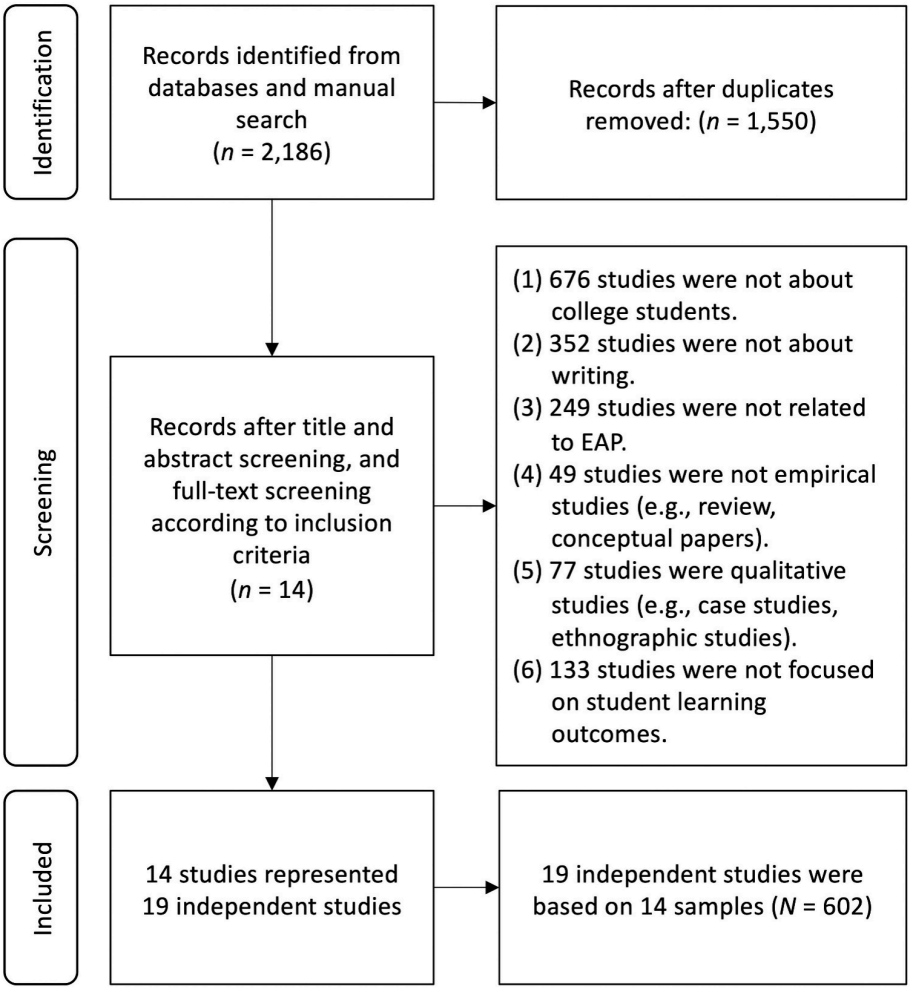


Figure 1. PRISMA flow chart for literature search (Page et al., 2021).

Coding procedure

As the first step of the coding, the authors reviewed the 14 studies to appraise their quality. We followed Jarde et al.'s (2012) suggested six domains of study design as reported in Table 2. First, 'representativeness' was used to check if the target samples could represent the population of our interest: college students registered for EAP writing programmes. Second, 'selection' was used to check if a sample selection process would bias the dataset construction. Third, 'measurement' was used to check if a study included appropriate measurements to assess target outcome variables. Fourth, 'data collection' was used to check if the data collection process was affected by any threats or problems when collecting the data. Fifth, 'statistics and data analysis' was used to check if appropriate data analysis approaches were used to compute accurate statistics to answer the research questions. Sixth, 'funding' was used to check if there were any conflicts of interests due to the sources of funding affecting the quality of study. Overall, we found that all 14 studies satisfied all but the 'selection' criterion, as they had convenience samples instead of including a random sampling approach; we concluded that every study had high qualities to be included in the current meta-analysis (see Appendix A for more detailed information about these six criteria).

Table 2. Quality appraisal of the 14 studies (adapted from Jarde et al., 2012).

Studies	Rep.	Sel.	Meas.	D.col.	Stat.	Fund.
Alghizzi and Alshahrani (2020)	√	–	√	√	√	√
Bestgen and Granger (2014)	√	–	√	√	√	√
Bulté and Housen (2014)	√	–	√	√	√	√
Crossley and McNamara (2014)	√	–	√	√	√	√
Crossley et al. (2016)	√	–	√	√	√	√
Crosthwaite (2016)	√	–	√	√	√	√
Crosthwaite and Jiang (2017)	√	–	√	√	√	√
Crosthwaite (2018)	√	–	√	√	√	√
Henry and Roseberry (1998)	√	–	√	√	√	√
Ishikawa (1995)	√	–	√	√	√	√
Mazgutova and Kormos (2015)	√	–	√	√	√	√
Park (2015)	√	–	√	√	√	√
Polio et al. (1998)	√	–	√	√	√	√
Shaw and Liu (1998)	√	–	√	√	√	√

Note. Rep. = Representativeness, Sel. = Selection, Meas. = Measurement, D.col. = Data collection, Stat. = Statistics and data analysis, Fund. = Funding.

After appraising the quality, we reviewed the 14 studies and found a complex structure among them. There were three cases, as illustrated in Figure 2. For example, Case #1: there was a study (the left diagram in Figure 2), which had two samples (independent studies), and each sample was analysed through two measurements, producing two ESs for each sample. For

example, Mazgutova and Kormos' (2015) study included two separate learner groups: CEFR C1 learners (n = 25) and CEFR B2 learners (n = 14). The diagram at the centre of Figure 2 illustrates Case #2: three studies that analysed one sample. Each study exhibited different measurements, but all the calculated ESs were actually from the same sample, meaning that there were six ESs. For example, three studies (Crosthwaite, 2016, 2018; Crosthwaite & Jiang, 2017) all analysed the same corpus (i.e. generally referring to a large body of linguistic data, and a collection of written essays from the same sample in this case), called the Hong Kong University learner corpus. The right diagram in Figure 2 presents Case #3, which is a conventional data structure in which a study had one sample, with ESs being associated with multiple measurements. As marked at the very left side of Figure 2, we found a two-level structure across these different data structures, where ESs are at Level 1, and samples at Level 2.

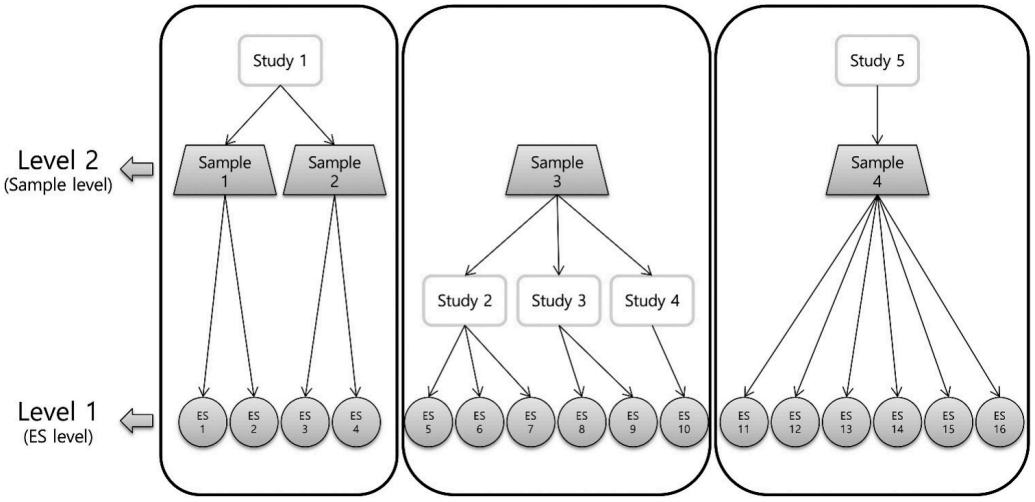


Figure 2. Structure of the data set with examples.

How the identified 14 studies were categorised into 14 samples after considering the three different data structures is represented in Table 3. Among the identified 14 studies, two had one sample each (Alghizzi & Alshahrani, 2020; Shaw & Liu, 1998), and seven were based on two samples (four on the Michigan State University corpus: Bestgen & Granger, 2014; Bulté & Housen, 2014; Crossley et al., 2016; Crossley & McNamara, 2014; three on the Hong Kong University corpus: Crosthwaite, 2016, 2018; Crosthwaite & Jiang, 2017). The remaining five studies (Henry & Roseberry, 1998; Ishikawa, 1995; Mazgutova & Kormos, 2015; Park, 2015; Polio et al., 1998) had two samples each. Collectively, we concluded that the 14 studies were based on 14 samples (N = 602).

Table 3. Identifying 14 samples from 14 studies.

Studies	Samples			Remarks
Alghizzi and Alshahrani (2020)	→	Alghizzi & Alshahrani (2020) (*n* = 16)		EAP in Saudi Arabia
Bestgen and Granger (2014) Bulté and Housen (2014) Crossley and McNamara (2014) Crossley et al. (2016)	→	Michigan State University learner corpus (2014) (*n* = 57)		EAP in the US
Crosthwaite (2016) Crosthwaite and Jiang (2017) Crosthwaite (2018)	→	Hong Kong University learner corpus (2016) (*n* = 86)		EAP in the HK
Henry and Roseberry (1998)	→	Henry and Roseberry (1998)[1]	Genre-based teaching (*n* = 17)	EAP in Brunei
	→	Henry and Roseberry (1998)[2]	Traditional EAP teaching (*n* = 17)	
Ishikawa (1995)	→	Ishikawa (1995)[1]	Narration-based class (*n* = 29)	EAP in Japan
	→	Ishikawa (1995)[2]	Question-based class (*n* = 28)	
Mazgutova and Kormos (2015)	→	Mazgutova and Kormos (2015)[1]	CEFR C1 learners (*n* = 25)	EAP in the UK
	→	Mazgutova and Kormos (2015)[2]	CEFR B2 learners (*n* = 14)	
Park (2015)	→	Park (2015)[1]	Extensive reading-based EAP (*n* = 44)	EAP in the US
	→	Park (2015)[2]	Traditional EAP teaching (*n* = 40)	
Polio et al. (1998)	→	Polio et al. (1998)[1]	Traditional EAP teaching (*n* = 34)	EAP in the US
	→	Polio et al. (1998)[2]	Revision enhanced EAP teaching (*n* = 31)	
Shaw and Liu (1998)	→	Shaw & Liu (1998) (*n* = 164)		EAP in the UK

Note. As can be seen in the table, it should not be mistakenly interpreted that the samples are matched one-on-one with each of the studies, although the number of the sample is coincidentally identical to that of the studies.

Table 4. Detailed information of the 14 samples across five moderators.

Samples	English proficiency	Target linguistic dimensions	Instructional conditions	Duration	Contexts
Alghizzi and Alshahrani (2020)	High	CAF measures	Regular EAP	1 semester	EFL
Michigan State University learner corpus (2014)	Intermediate	CAF measures + Other measures	Regular EAP	1 semester	ESL
Hong Kong University learner corpus (2016)	High	CAF measures + Other measures	Regular EAP	1 semester	ESL
Henry and Roseberry (1998)[1] Henry and Roseberry (1998)[2]	High	Other measures	Treated EAP Regular EAP	3 weeks	ESL

Samples	English proficiency	Target linguistic dimensions	Instructional conditions	Duration	Contexts
Ishikawa (1995)[1] Ishikawa (1995)[2]	Low	CAF measures	Regular EAP	1 semester	EFL
Mazgutova and Kormos (2015)[1] Mazgutova and Kormos (2015)[2]	High Intermediate	CAF measures	Regular EAP	4 weeks	ESL
Park (2015)[1] Park (2015)[2]	Intermediate	Other measures	Treated EAP Regular EAP	1 semester	ESL
Polio et al. (1998)[1] Polio et al. (1998)[2]	Intermediate	CAF measures	Treated EAP Regular EAP	1 semester	ESL
Shaw and Liu (1998)	Intermediate	Other measures	Regular EAP	2 ~ 3 months	ESL

Detailed information about the samples is presented in Table 4. We found that these were different across a total of five identified moderator variables.

First, we found that each college EAP programme was presented for students with different English proficiency levels. A total of five samples (36%) concerned students with high proficiency, while seven (50%) and two (14%) related to intermediate and low proficiency, respectively.

Second, we found that each sample examined unique sets of target linguistic dimensions to evaluate their students' EAP writing performance based on the CAF (Ellis & Barkhuizen, 2005; Skehan, 1998; Wolfe-Quintero et al., 1998) or other measures. For example, nine samples focused on improvements or changes in CAF measures of their students' writing performance, and seven used combinations of different measurements, such as holistic assessments, cohesiveness, and styles.

Third, we found that a total of 11 samples (79%) were from a traditional learning condition (i.e. a business-as-usual condition), whereas the remaining three (21%) were based on an intervention condition, in which institutions adopted a novel approach.

Fourth, the course duration for most of the college EAP programmes (9 samples; 64%) was one semester, while the remaining five (36%) came from EAP programmes shorter than a semester.

Fifth, we found that the programmes came from different contexts around the world. A total of 11 samples (79%) came from ESL contexts (e.g. the US, the UK), whereas three (21%) came from EFL ones (e.g. Saudi Arabia and Japan).

Effect size calculation

To estimate the standardised degree of the improvements in English writing through college EAP programmes, we computed ESs using the pretest and posttest data of each programme. Provided that the included samples had overall small sample sizes, we used unbiased *d* (also known as Hedges' *g*; Hedges & Olkin, 1985) for our ES calculation. In other words, we multiplied an additionally computed correction factor [*J*; equation (3)] to Cohen's *d* to eliminate any bias coming from small sample sizes (e.g. $n < 50$). The equations for ESs [equation (1)] and their standard errors [equation (2)] are presented below.

$$ES_n = J_{Correction\ Factor} \times \frac{Mean_{post-test} - Mean_{pre-test}}{\sqrt{\left[\frac{\left(n_{post-test}-1\right)SD^2_{post-test}+\left(n_{pre-test}-1\right)SD^2_{pre-test}}{n_{post-test}+n_{pre-test}-2}\right]}} \quad (1)$$

$$SE_n = J_{CorrectionFactor} \times \sqrt{\frac{1}{n_{post-test}} + \frac{1}{n_{pre-test}} + \frac{\text{Cohen's } d^2}{2 \times \left(n_{post-test} + n_{pre-test}\right)}} \quad (2)$$

$$J_{Correction\ Factor} = 1 - \frac{3}{\left\{4 \times \left(n_{post-test} + n_{pre-test}-^2\right) - 1\right\}} \quad (3)$$

As shown in Table 5, we computed a total of 175 ESs based on 14 samples. For an EAP programme in Saudi Arabia in Alghizzi and Alshahrani (2020), we computed four ESs based on one sample, and all the ESs were based on CAF measures. The Michigan State University (MSU) learner corpus, which was analysed in four different studies, had a total of 50 ESs based on multiple types of measures, such as holistic assessments, complexity measure, cohesion measure, and style measure. The Hong Kong University learner corpus, analysed in three studies, had 16 ESs related to style and accuracy measures. Based on the EAP programme in Brunei with two different samples, Henry and Roseberry (1998) had six ESs based on style measure. Similarly, Ishikawa (1995) in Japan, Mazgutova and Kormos (2015) in the UK, Park (2015) in the US, and Polio et al. (1998) in the US reported their EAP programmes with two different samples that contributed 48 ESs based on CAF measures, with 20 ESs based on complexity measure, 12 on holistic assessments and style measure, and eight on accuracy measure. Lastly, Shaw and Liu (1998) in the UK offered 11 ESs based on style measure. The data collection sheet, which includes more detailed information about the actual coding procedure as well as the calculated ES estimates, is included in Appendix B.

Table 5. Number of effect sizes for 14 samples.

Samples	**Effect size estimates**				
	Total	**Complexity**	**Accuracy**	**Fluency**	**Other measures**
Alghizzi and Alshahrani (2020)	4	3	1		

Samples	Effect size estimates				
	Total	**Complexity**	**Accuracy**	**Fluency**	**Other measures**
Michigan State University learner corpus (2014)	50	6			
		13			Holistic (3)
		11			Holistic (2)
					Cohesion (15)
Hong Kong University learner corpus (2016)	16				Style (5)
					Style (4)
			7		
Henry and Roseberry (1998)[1]	3				Style (3)
Henry and Roseberry (1998)[2]	3				Style (3)
Ishikawa (1995)[1]	24	11	12	1	
Ishikawa (1995)[2]	24	9	11	4	
Mazgutova and Kormos (2015)[1]	10	10			
Mazgutova and Kormos (2015)[2]	10	10			
Park (2015)[1]	6				Holistic (1), Style (5)
Park (2015)[2]	6				Holistic (1), Style (5)
Polio et al. (1998)[1]	4		4		
Polio et al. (1998)[2]	4		4		
Shaw and Liu (1998)	11				Style (11)
Total	175				

Note. Numbers in parentheses for effect size estimates indicate the number of effect sizes calculated for each study

Multilevel meta-analysis plan

As previously mentioned, we computed a total of 175 ESs. However, these did not simply come from a hierarchical structure unlike many conventional ESs in the field (e.g. one-level data structure: ESs from studies, Boulton & Cobb, 2017; two-level data structure: ESs nested in studies, Lee et al., 2019; three-level data structure: ESs nested in studies derived from research projects, Lee et al., 2020). In Figure 2, we previously depicted three examples of the complex structure of our data set. To take this data structure into consideration by separating different levels of variance components (see Hox et al., 2010 for theoretical background of this approach), we used a multilevel modelling approach for further data analyses. In this way, we could avoid the independence issue such as that the MSU learner corpus, which was analysed in four studies, is not weighted as being four times more important than a sample with one study. To this end, we employed a multilevel regression analysis with the ESs as the dependent variable using STATA 14 and HLM 7 (see Lee et al., 2019 for practical guidance for such an approach) to answer the first research question.

To answer the second research question, we adopted Plonsky and Oswald's (2014) ES interpretation benchmark approach, in which ESs at the 25th, 50th, and 75th percentiles were designated as benchmarks. However, in view of the complex multilevel structure of our ESs, instead of using percentiles, we decided to compute mean ES estimates after controlling for the moderator variables for the following three sections: (1) between the 0th and 50th, (2) between the 25th and 75th, and (3) between the 50th and 100th percentiles. In using mean estimates, there is as a possibility of outliers in the data set; nonetheless, just picking three numbers in order (i.e. the 25th, 50th, and 75th percentiles) without considering ES backgrounds (ESs are from different EAP programmes), nested structures (ESs are nested in samples), as well as sampling errors (ES are based on different sample sizes) would be less desirable.

To answer the third research question, we conducted a meta-regression where we computed mean ES estimates for each value of the moderator variables and checked the coefficients. In this way, we can compute the mean ES estimates for each moderator variable while controlling for the other moderator variables and how these estimates were different from the mean ES estimates computed by each value.

Results

Overall average effect size estimate

To graphically display the computed ESs, a forest plot is widely used. However, due to limitations of space, for all 175 ESs we illustrated a plot for the overall ES estimates at the sample level (k = 14) in Figure 3. For this reason, this figure should be read only for visualising the general trend of the computed ESs. The forest plot shows that eight out of 14 samples (57%) included zeros in their 95% confidence intervals, indicating that these studies had null intervention effects of EAP writing programmes on students' writing skills. The remaining (6 out of 14; 43%) had statistically significant effect size estimates, ranging from 0.02 to 1.21, and this indicated that college EAP writing programmes had an overall positive impact on improving students' writing skills.

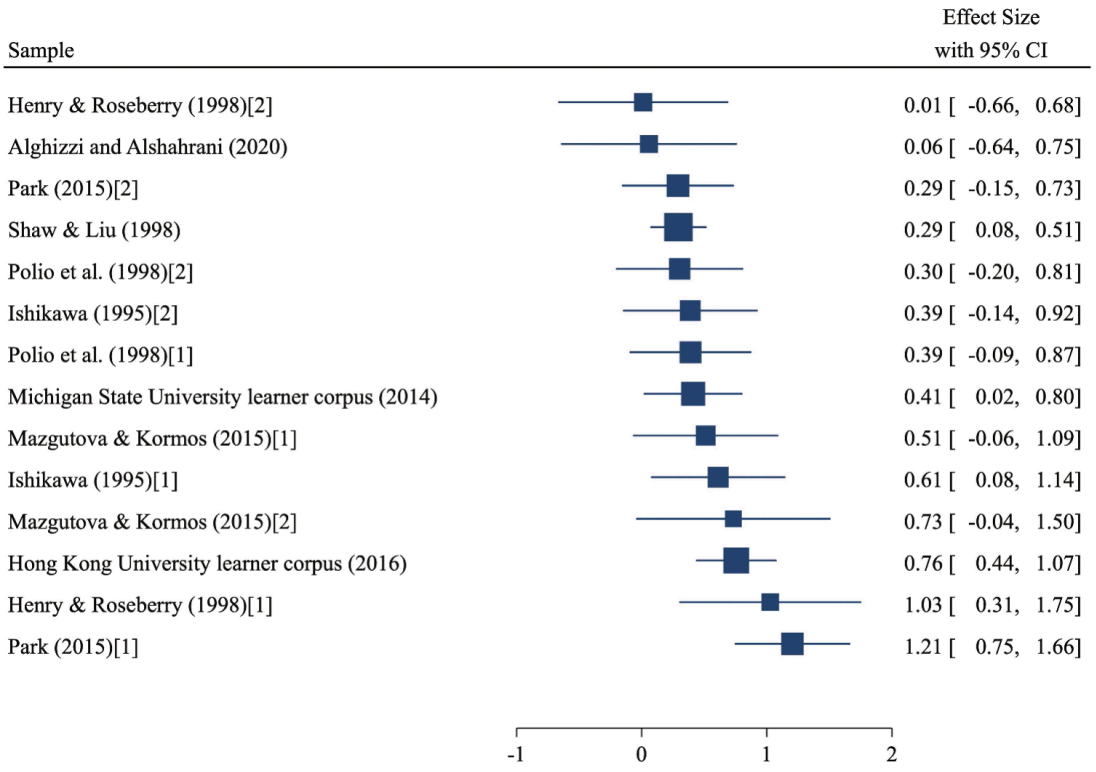


Figure 3. Forest plot for the overall effect size estimates at the sample level.

The results of our two-level regression models for overall average ES estimates with an inclusion of moderator variables are reported in Table 6. Results indicated that the overall average ES was 0.45 (*SE* = .07, $p < .001$) based on 175 ESs based on 14 samples, keeping moderator variables at their means. According to the most widely accepted Cohen's ES interpretation, where 0.2, 0.5, and 0.8 are respectively considered small, medium, and large, the overall estimate of the degree of the improvements in students' English writing through college EAP programmes could be seen as *small*. However, according to Plonsky and Oswald's (2014) ES interpretation benchmarks, in which for within-group comparisons (i.e. pretest–posttest designs), 0.6, 1.0, and 1.4 are considered small, medium, and large ESs, respectively, this overall estimate could be seen as either small or marginal. In the context of L2 research, this figure was lower than the 25th percentiles of interventions (Plonsky & Oswald, 2014).

Table 6. Overall mean effect size (ES) estimates for college EAP programmes.

No. of ESs	No. of samples	Mean ES	*SE*
175	14	0.45***	(0.07)

Note. To take differences across EAP college programmes into account, the mean ES estimates were controlled for the following five moderator variables: (1) learners' English proficiency levels, (2) target linguistic dimensions, (3) instructional condition, (4) duration, and (5) contexts. *** $p < .001$.

Effect size interpretation benchmarks

Following Plonsky and Oswald's (2014) suggestion, we further computed our own ES interpretation benchmarks for this specific discipline. For small, medium, and large benchmarks, we calculated mean ES estimates between the 0th and 50th, the 25th and 75th, and the 50th and 100th percentiles. As shown in Table 7, the results indicated that 0.3, 0.5, and 0.7 are statistically significant ($p < .001$) benchmarks for small, medium, and large, respectively. Thus, these benchmarks may gauge how large the degree of the improvements in English writing through college EAP programmes is. If a mean ES estimate is less than 0.3, we could consider this *marginal*. If a mean ES is larger than 0.3 and smaller than 0.5, the size of the success is *small*, while it should be larger than 0.5 and smaller than 0.7 to be considered *medium*. Its success would be considered *large* if its mean ES estimate is greater than 0.7. Each mean ES estimate came from the same number of ESs, and their intervals were smaller than those of Plonsky and Oswald's (2014) ES interpretation benchmarks (i.e. 0.6, 1.0, 1.4).

Analysis of moderator variables

The results of our simple- and meta-regression analyses for moderator variables are reported in Table 8. First, regarding students' English proficiency, the results of the simple regression revealed that the college EAP programmes had a medium-sized positive impact on improving students' writing across different English proficiency levels ($d \approx .50$). However, the results of the meta-regression indicated that the college EAP programme for writing had a large-sized impact for students with low English proficiency ($d = .81$, $SE = .28$, $p < .01$) and a small-sized impact for those with high ($d = .39$, $SE = .14$, $p < .01$) and intermediate ($d = .29$, $SE = .14$, $p < .05$) proficiency. Nevertheless, we found that this seemingly large ES difference between these values was not statistically significant.

Second, regarding the target linguistic dimensions, the results of both simple- and meta-regression revealed that the college EAP programmes had a larger impact on improving accuracy than on complexity and fluency. In particular, the results of the meta-regression indicated that the programmes had a small-sized impact on improving the accuracy ($d = .41$, $SE = .09$, $p < .001$) and complexity ($d = .31$, $SE = .09$, $p < .001$) of students' EAP writing and a marginal impact on improving their fluency ($d = .18$, $SE = .18$, $p > .05$), which was not statistically significant. For the other measures, we found that the programmes had a medium-sized impact ($d = .68$, $SE = .08$, $p < .001$) on improving overall writing quality, cohesiveness, and/or styles of students' EAP writing. Regarding the contrast among the CAF measures, we found that the ES differences across different dimensions were not statistically significant. It

should be noted here that we coded measurements other than the CAF framework as a composite categorical value to secure sufficient statistical power and to allow us to interpret the results according to appropriate theoretical perspectives.

Table 7. Effect size interpretation benchmarks for EAP programmes.

	Small (≈25th: mean ES between 0th and 50th)	**Medium** (≈50th; mean ES between 25th and 75th)	**Large** (≈75th; mean ES between 50th and 100th)
Benchmarks	0.3	0.5	0.7
Mean ES	0.25*** (0.03)	0.46*** (0.08)	0.71*** (0.03)

Note. Mean ES estimates are statistically adjusted values after controlling for the following five moderator variables: (1) learners' English proficiency levels, (2) target linguistic dimensions, (3) instructional condition, (4) duration, and (5) contexts. *** $p < .001$.

Table 8. Simple- and meta-regression for moderator variables.

Moderator variable	**No. of ESs**	**No. of samples**	**Simple-regression**		**Meta-regression**		
			Mean ES	***SE***	**Mean ES**	***SE***	**Contrast**
1. English proficiency							
(1) High	36	5	0.54a**	(0.15)	0.39**	(0.14)	(3) ≈ (1) ≈ (2)
(2) Intermediate	91	7	0.51***	(0.12)	0.29*	(0.14)	
(3) Low	48	2	0.49***	(0.08)	0.81**	(0.28)	
2. Target linguistic dimensions							
(1) Complexity	73	6	0.39***	(0.06)	0.31***	(0.09)	(2) ≈ (1) ≈ (3)
(2) Accuracy	39	6	0.48***	(0.08)	0.41***	(0.09)	
(3) Fluency	5	2	0.28*	(0.14)	0.18	(0.18)	
(4) Other measures	58	7	0.63***	(0.15)	0.68***	(0.08)	
3. Instructional condition							
(1) Intervention condition	13	3	0.90***	(0.00)	0.80***	(0.17)	(1) > (2)
(2) Business-as-usual condition	162	11	0.44***	(0.06)	0.42***	(0.08)	
4. Duration							
(1) Shorter than 1 semester	37	5	0.48***	(0.13)	0.43**	(0.14)	(2) ≈ (1)
(2) 1 semester	138	9	0.50***	(0.10)	0.46***	(0.09)	
5. Context							
(1) ESL (e.g. the US, the UK)	123	11	0.53***	(0.09)	0.53***	(0.14)	(1) ≈ (2)
(2) EFL (e.g. Japan)	52	3	0.42***	(0.11)	0.26	(0.25)	

a $p < .05$, ** $p < .01$, *** $p < .001$.

With respect to the three remaining moderator variables, both the simple- and meta-regression revealed similar estimates; that is, on average, EAP programmes with an intervention condition ($d = .80$, $SE = .17$, $p < .001$) had larger programme effects (ES difference = .37, $SE = .18$, $p < .05$) than those with the traditional condition ($d = .42$, $SE = .08$, $p < .001$). In terms of the durations, the results indicated that EAP programmes conducted for one semester or shorter (e.g. one or two months) had statistically similar programme effects ($p > .05$). In terms

of the context, we found that the EAP programmes conducted in ESL contexts (d = .53, SE = .14, p < .001) had a larger programme effect than those conducted in EFL ones (d = .26, SE = .25, p > .05), although the ES difference was not statistically significant.

Discussion

Assessing the success of EAP programmes for writing and its implications

The present meta-analysis was based on 175 ESs calculated from 14 samples. The overall average ES estimate was .45, suggesting that the degree of the improvements in English writing through college EAP programmes could be considered *marginal* in view of Plonsky and Oswald's (2014) ES interpretation benchmarks for within-group comparisons when compared to other L2 research. However, this figure could also be interpreted as *small* according to Cohen's traditional ES interpretation benchmarks, where .20, .50, and .80 are for small, medium, and large, respectively. In other words, we can conclude that English learners could expect to develop their writing proficiency (or performance), albeit to a limited extent, by taking college EAP programmes. We believe that such results come from the combined effects of the task-based approach (Skehan, 1996), the genre approach to academic writing (Hyland, 2007, 2008), and instructors' feedback on learners' writing (Bitchener & Ferris, 2012; Ferris, 2002), which are all taken as elements of EAP programmes.

Further, this marginal ES, which can be seen as small according to different interpretation benchmarks, also invites other possible explanations. For example, in view of the multi-dimensional characteristic of L2 writing (Silva & Brice, 2004), taking one or two EAP courses may not be sufficient for EFL or ESL learners to reach the required level of English writing proficiency. This finding somewhat agrees with the results of a meta-analysis on the effectiveness of L2 instruction (Plonsky, 2011), in which strategy instruction was more effective for skills such as speaking, reading, pronunciation, and vocabulary than for writing. Thus, taken together, this overall average ES of the degree of the improvements in English writing through college EAP programmes suggests the slow-developing nature of writing skills, and also a need for more tailored ES interpretation benchmarks to interpret the results.

Our attempt to provide ES interpretation benchmarks for EAP writing research, so that those concerned (e.g. instructors in these programmes and researchers on this topic) could evaluate the success of their own programmes, was therefore a necessary and meaningful step. Our result indicated that, for the success of college EAP programmes, .3, .5, and .7 are statistically significant benchmarks for small, medium, and large effects, respectively, which were relatively lower than Plonsky and Oswald's (2014) ES interpretation benchmarks (i.e. .6, 1.0, 1.4). For writing skills, it takes more time and effort to develop to a high degree in comparison

with other linguistic skills (Polio et al., 1998). Although a number of studies on the success and effectiveness of college EAP programmes are needed, we believe that our benchmarks could serve as useful guidelines for EAP researchers and teaching practitioners for evaluation purposes.

Evidence-based suggestions to effectively manage EAP programmes for writing

Turning to the results related to the third research question, students' English proficiency and target linguistic dimensions influenced the degree of the improvements in English writing through college EAP programmes in significant and complex ways. Regarding the first moderator variable, the meta-analysis identified a noteworthy effect of the collected EAP programmes for writing on students of low proficiency, such that these students (d = .81) benefited more than their higher proficiency counterparts (d = .29 ~ .39), with the ESs being large and small in terms of our ES benchmarks. Such a rejection of a so-called threshold level effect may have come from the curriculum and fine-tuned writing instruction of the programmes geared towards students with lower levels of English proficiency, in view of the clear recognition that 'EAP is simply too hard for students at lower levels of English proficiency who need to acquire a "general English" suitable for all contexts before they can study the complexities of academic discourse' (Hyland, 2016, p. 18).

In terms of the CAF framework (Ellis, 2003; Ellis & Barkhuizen, 2005; Skehan, 1998; Wolfe-Quintero et al., 1998), as part of another moderator variable of target linguistic dimensions, accuracy demonstrated the largest growth out of the three components (i.e. complexity, accuracy, fluency). This finding may be a rather unexpected one in that improvement in accuracy is often not an explicit goal of these programmes; however, it should be noted that 'EAP teachers do spend many hours proof-reading and correcting such errors in their student's written production via a variety of online and offline feedback mediums' (Crosthwaite, 2018, p. 316), and also some common grammatical errors are frequently addressed at the class level (Storch & Tapper, 2009). Additionally, as an important feature of these courses, instructors' feedback on students' written assignments (e.g. Ishikawa, 1995; Mazgutova & Kormos, 2015) may have contributed to the improvements of accuracy to a greater extent than those of complexity and fluency in these students' writing. However, we take this finding as a positive sign that college EAP programmes could contribute to the improvements of students' writing in all three components of the CAF framework, with accuracy being improved slightly more than complexity and fluency.

Regarding the 'other measures' category, holistic assessments, cohesion and/or style exhibited an overall medium ES of college EAP programmes (d = .68). As we coded this categorical value as a composite of different measurements due to practical reasons (i.e. to secure enough

statistical powers and to ensure theoretically appropriate interpretations), it is not possible to suggest a detailed explanation of how each measurement plays a role in interpreting the degree of the improvements in students' English writing through EAP programmes. Regarding the cohesiveness- and style-related measurements, the improvements in these dimensions may be attributed to the characteristics of EGAP programmes, which are oriented towards teaching 'general academic English register' and 'formal, academic style proficiency in language use' (Jordan, 1997, p. 3). It is also noteworthy that students registered in EGAP courses often also take 'main degree subjects' (Crosthwaite, 2016), which may address discipline-specific registers, albeit not in depth. It is possible that taking these two types of courses simultaneously may have contributed to the developments in these dimensions, as well as holistic assessment scoring, which may have reflected raters' overall impression of the growth of different dimensions in writing.

Regarding the rest of the moderator variables, the categories in duration and contexts were not significantly different from each other, although programmes in ESL contexts showed a larger ES than those in EFL contexts. However, compared to the business-as-usual one, the intervention condition demonstrated a significantly larger ES, highlighting its pedagogical value. The intervention conditions included in the sampled studies range from genre-based teaching in Henry and Roseberry (1998), extensive reading condition in Park (2015), to the revision-enhanced condition in Polio et al. (1998). Due to a small number of ESs and empirical studies, we cannot statistically compare the relative effects of each intervention condition; however, the results of this moderator analysis suggest that exercising such intervention conditions in EAP programmes may be able to accelerate the development of English writing, which progresses slowly by nature. Future studies on college EAP programmes are encouraged to explore this moderator to contribute to the values of these instructional conditions.

Conclusion

This meta-analysis was conducted to provide an estimate of the degree of the improvements in students' English writing through college EAP programmes as well as to examine how a range of moderator variables influence the degree of the improvements through such programmes. A quality appraisal was included as part of our meta-analytic efforts, and it showed the high qualities of the constituent studies, providing some evidence regarding the trustworthiness of the findings of this meta-analysis as a whole. The results revealed that these programmes did contribute to the improvements in students' English writing, but more so for low proficiency learners, and in terms of certain linguistic dimensions such as accuracy and complexity. Furthermore, an intervention condition realised more gains in writing than a business-as-usual

condition. We also proposed ES benchmarks for students' improvements in English writing through EAP programmes, which showed different ranges from those of Plonsky and Oswald (2014).

However, this study is not without its limitations. First, the number of primary studies included in the meta-analysis was rather small (k = 14), which has made the interpretation of the analyses rather difficult. Second, our results are largely dependent on the studies that adopted a within-group design. A rather heavy reliance on such a methodological design reflects a practical constraint that it is difficult to sample two groups, each of which is allocated to either an EAP or a non-EAP course (Helms-Park et al., 2007). While we have made efforts to overcome these limitations through rigorous statistical procedures for data analysis, it does not downplay the need for a greater number of empirical studies on the success and effectiveness of college EAP programmes for linguistic developments and those based on different methodological designs.

Apart from more studies on this issue, we also call for more research that deals with the effects of various course features of EAP programmes on the development of English writing. One possible candidate is the relative (or combined) effects of instructors' online and offline feedback (Crosthwaite, 2018), which is now widely possible through the adoption of online-based education platforms. It may be also worthwhile to investigate the role and effects of the flipped model in the context of college EAP programmes, which has started to attract attention in the field of English writing pedagogy (e.g. Leis et al., 2015). It is expected that more studies with a focus on diverse features of programmes would provide more detailed pedagogical implications for those involved in teaching writing in college EAP programmes. Our meta-analysis study, we believe, has taken a concrete and important step in such a direction.

Disclosure statement

No potential conflict of interest was reported by the authors.

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